REGIME-AWARE ML · 2026 · PROJECT LABORATORY PRESENTATION

Regime-Aware Machine Learning for Technical Patterns on SPY

Event-based learning on technical chart patterns, triple-barrier labelling, and a profitability-versus-classification trade-off study on **SPY** (S&P 500 ETF · U.S. large-cap stocks), daily bars, **2010 → 2025**.

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Why financial ML refuses to behave.

Markets are not stationary. Over ~4,000 SPY daily bars (2010 → 2025), feeding every bar to a classifier drowns the few rare, high-information moments under noise. **Event-based learning** samples only the moments that geometrically *matter*.

01 **Bars are 97 % noise**

Daily returns are dominated by microstructure and macro shocks the model cannot see.

02 **Patterns are humans' prior**

Eighty years of technical analysis literature says these shapes are where traders actually act.

03 **One row = one decision**

The detector decides *when*. The classifier only chooses *what*: long, short, or skip.

DATASET

~4,000	137
SPY DAILY BARS	EVENTS · 3.4 %

Research question. Can a geometry-conditioned classifier, labelled by a triple barrier, deliver out-of-sample profitability while avoiding the leakage that haunts daily-bar ML?

Four detectors, one event book.

Each detector outputs a completion bar plus slopes, intercepts and touch statistics. The smallest set that closes the 2-D pattern algebra: horizontal level, sloped level, converging pair, repeated extrema.

Support / Resistance

n = 42. Horizontal level from prior swing extrema. Fires when close re-enters **±0.3·ATR**[†] of the level.

Channels

n = 12. Parallel trendlines fitted by least-squares (`np.polyfit`), containment ≥ 0.70.

Triangles

n = 22. Converging lines via `linregress`, **|r| ≥ 0.85**, containment ≥ 0.80.

Multi top / bottom

n = 63. ≥ 2 swing extrema clustered at a common level. Dominant family on the SPY tape.

EVENT BOOK

137	94
TOTAL DETECTED	USED IN TRAINING

Triangles (22) and Channels (12) excluded from the NB12 training set; 94 events carry valid triple-barrier labels.

WHY THESE FOUR

The **smallest set that closes the 2-D pattern algebra**. H&S, flags and wedges decompose into combinations of these.

Reading the figures. The ◆ yellow diamond marks the signal bar.

The signal bar (yellow diamond) is the only row.

Features look backward from `t`. Labels look forward from `t + 1`. The two windows never overlap.

L/S/N

DETECT → **SIGNAL BAR** → **FEATURES** → **TRIPLE BARRIER** → **LABEL**

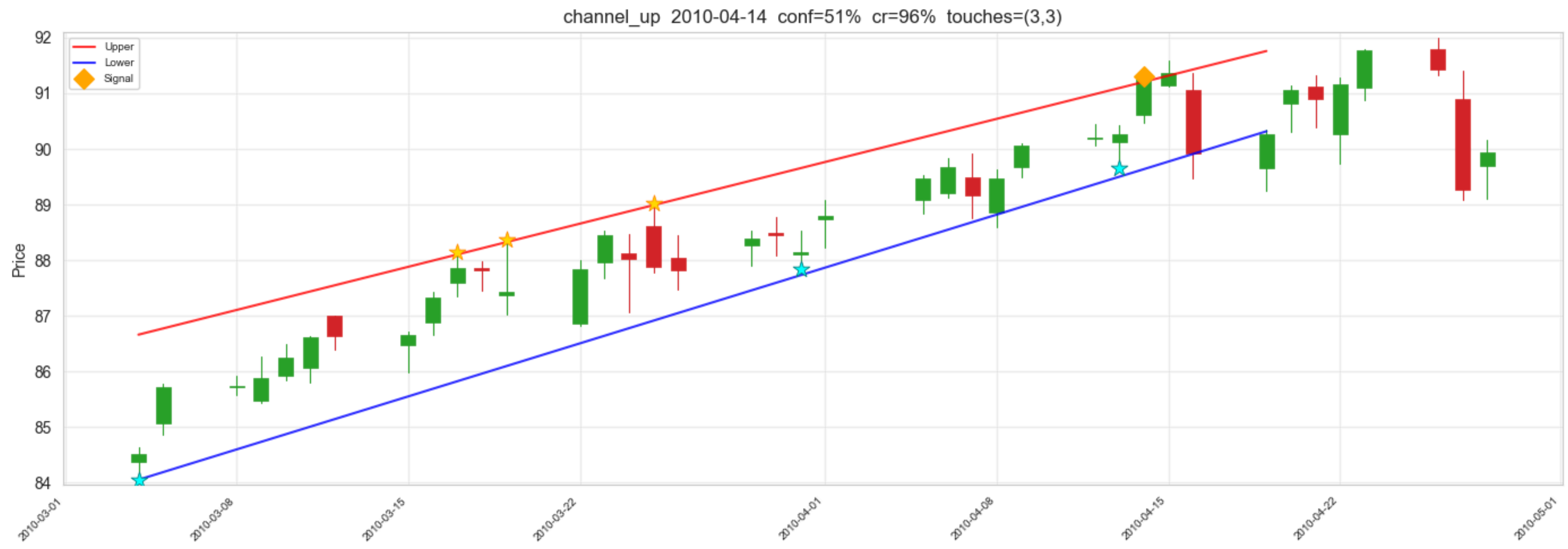

Fig 4.1. Ascending channel completing 2010-04-14. The yellow diamond is the signal bar.

BACKWARD WINDOW

33 indicators + 12 geometry features at the event close. **No look-ahead.**

SIGNAL BAR · T

One row per event. Pattern formed over 20–75 bars; only the **completion close** enters the matrix.

FORWARD WINDOW

First of `{TP, SL, max_holding}` hit → label is **long**, **short** or **no-trade**.

Six families, one signal bar.

Around 45 columns drawn from price, volume and the fitted pattern geometry. Standardised on the train fold only; the scaler never sees validation or test events.

T
Trend
SMA distances, MA spreads

V
Volatility
ATR, BB width, %B

M
Momentum
Returns, RSI, MACD

Q
Volume
OBV, vol σ & ratio

G
Geometry
Slopes, touches, contain.

B
Binary
BB breach, SMA cross, RSI

TOP FEATURES · ANOVA F-STATISTIC

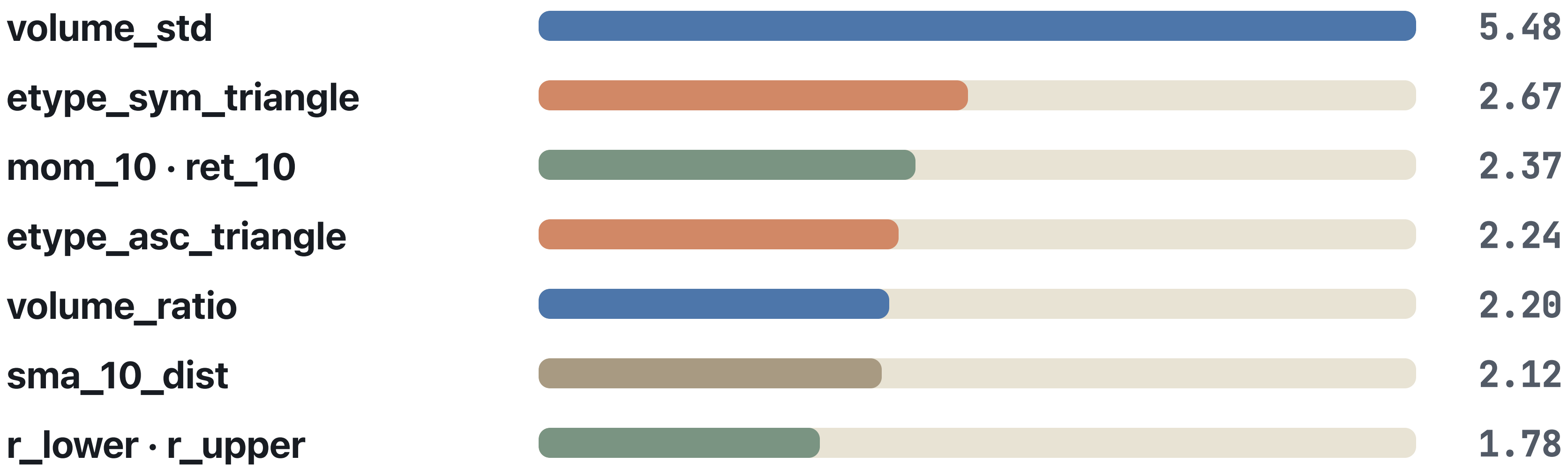

Fig 5.1. Only *volume_std* clears the 1 % significance threshold. The model relies on **interactions**, not individual features.
Note. Computed on the full pre-exclusion set; triangle `etype_` dummies are dropped in the actual training matrix.

FEATURE RATIONALE

Trend & momentum, direction. Without them a long signal on a bear day looks identical to a short on a bull day.

Volatility & volume, regime indicators. Quiet tape vs noisy tape demand different decisions.

Geometry features, slopes, touches, fit error. Tight, well-formed channel vs loose one.

Binary signals, discrete, robust to scaling, anchor the ensemble when continuous features drift.

Removed (trend-leaking): raw ATR, absolute SMAs, event ATR and `obv_norm`. Kept normalised `atr_ratio`, `volume_ratio`, `volume_std` so the model cannot infer absolute price level.

Three barriers, one honest label.

Starting at the event bar, three barriers race for first contact. Take-profit at $+pt \cdot \sigma_{ATR}$, stop-loss at $-sl \cdot \sigma_{ATR}$, time barrier at `t + max_holding`. Whichever is hit first decides the label, the classical López de Prado (2018) recipe.

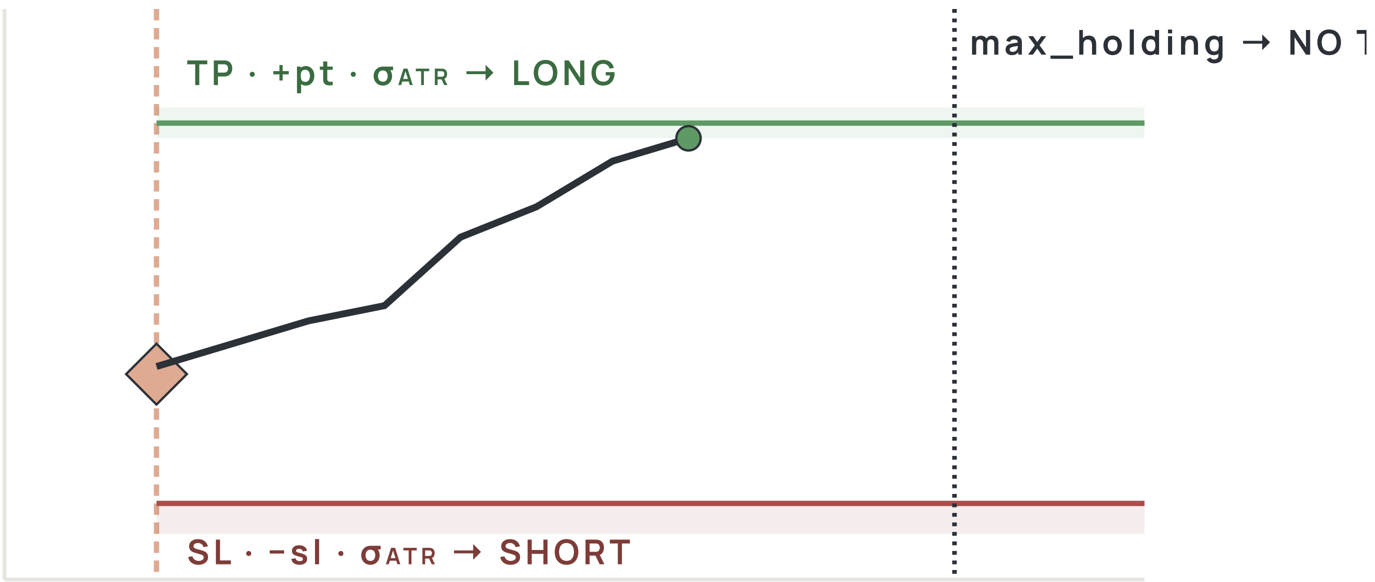

Fig 6.1. The triple-barrier method drawn at scale. Wins to the upside become LONG, losses become SHORT, time-outs become NO-TRADE.

HYPERPARAMETERS · GRID & OPTIMA

PARAM	GRID	BEST F1	BEST PROFIT
pt_mult · profit-take	1.0–3.0	2.0	2.0
sl_mult · stop-loss	1.0–3.0	1.5	3.0
max_holding · bars	5–20	10	20

Why optimised, not fixed. Labels *are* a hyperparameter, same model under different barriers learns different things. **100 configurations** swept.

0.654

BEST F1

pt=1.0 · sl=2.5 · mh=15

+18.7%

BEST PROFIT · BAGGING

pt=2.0 · sl=3.0 · mh=20 · ceiling +28.7 %

The disagreement is the finding. Max F1 ≠ max return. Page 10 returns to this.

Localise then trust.

Every detector is audited for geometric correctness and signal localisation before any classifier touches the data.

CHANNEL TOUCH STATISTICS · N = 12

DATE	TYPE	TOUCHES	CR
2010-04-14	channel_up	3 · 3	.96
2012-11-13	channel_down	2 · 4	.96
2014-02-13	channel_down	3 · 3	1.00
2016-04-26	channel_up	3 · 4	.97
2019-07-19	channel_up	4 · 4	1.00
2023-08-10	channel_up	2 · 3	1.00

All accepted channels meet ≥ 2/2 touches and ≥ 0.70 containment. **Median 0.99.**

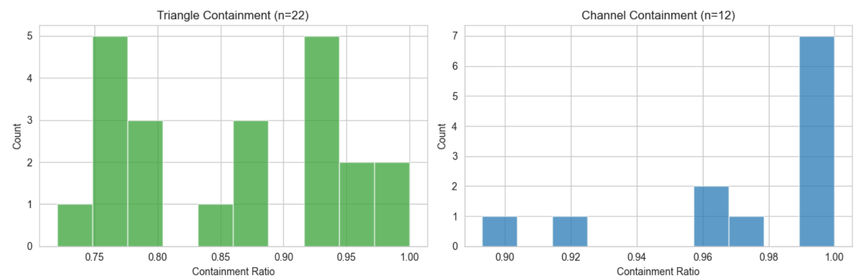

Fig 7.1. Containment ratios, triangles median 0.86, channels median 0.99. Geometry is empirically genuine.

Signal-bar logic · v2. The event bar is now the **last** bar in the pattern window, not the first. Features are evaluated at that close, removed the strongest source of look-ahead in v1.

QUALITY GATES · PER DETECTOR

DETECTOR	WINDOW	|R|	CONTAINMENT
Triangles	25 bars	|r| ≥ 0.85	≥ 0.80
Channels	25–55 bars	polyfit	≥ 0.70
S / R	rolling	–	±0.3·ATR band
Multi top / bot.	20-bar roll	–	slope-confirmed

Channels fit trendlines with `np.polyfit`; triangles use `linregress` on swing pivots. Tuning iterated from **1,067 → 137** events (26.5 % → 3.5 % of bars).

SUPERVISOR-DRIVEN IMPROVEMENTS

1. Localised signal at `t = event`
2. Removed trend-leaking features
3. Binary technical signals added
4. Touch events: +38 events
5. Walk-forward CV replaces single split

Random Forest and Bagging.

Both tree-based ensembles, robust on tabular data, well-suited when no single feature dominates. We train both and compare.

RANDOM FOREST

Standard sklearn ensemble with class-balanced weights. Splits on Gini.

n_estimators	200
max_depth	8
max_depth	8
class_weight	balanced
random_state	42

BAGGING CLASSIFIER

Bootstrap aggregation over shallow decision trees, the more profitable of the two.

base_estimator	DT(depth=8)
n_estimators	200
max_samples	1.0
max_features	1.0
random_state	42

END-TO-END PIPELINE

~4k
DAILY BARS

104
EVENTS

45
FEATURES

L/S/N
TRIPLE BARRIER

★
RF · BAGGING

Leakage controls. Standardiser fitted on the train fold; pattern parameters never use future bars; labels computed from t + 1 only.

Class imbalance. 94 labelled events in the NB12 pipeline. Distribution 38 L · 25 N · 31 S. Inverse-frequency weights, no resampling.

Baseline. Stratified random classifier. On the small test fold it edges ahead on accuracy and F1 (.447), a sign of small-sample variance, not signal.

Three checks against self-deception.

Chronological split for the final test, plus 5-fold event-CV on the train block. Folds are over **events**, not bars, no half-pattern leaks across folds.

TRAIN · 60 %	VAL · 20 %	TEST · 20 %
2010-04 → 2019-08 · 82 events	→ 2022-04 · 27	→ 2025-12 · 28

Fig 9.1. Chronological 60/20/20 split. Lineage: **137** detected → triangles + channels excluded → **94** events with valid triple-barrier labels.
plication →
94
with valid triple-barrier labels (10 dropped, insufficient forward window).

5-FOLD EVENT CV · PER-FOLD METRICS

FOLD	$F1_M$	ACC	CUM. RET
1	.412	.471	+0.084
2	.398	.500	+0.121
3	.456	.529	+0.142
4	.371	.412	-0.022
5	.488	.500	+0.176
$\mu \pm \sigma$	.425 ± .046	.482 ± .045	+.100 ± .076

Fold 4 is the harsh fold, 2020-Q1 events under COVID volatility. Removing it lifts mean F1 to .459.

F1 VS F-B · WHY BOTH MATTER

F1, harmonic mean of precision and recall. Class-symmetric.

F-β, β < 1 rewards precision (don't trade unless sure); β > 1 rewards recall (don't miss the move).

Trading costs are asymmetric, opportunity cost ≠ realised loss. Page 11 discusses this further.

What we don't claim. 94 events is small. We do not report individual-fold deltas as significant, we report distributions and let the test set adjudicate.

WALK-FORWARD VS K-FOLD

Walk-forward respects time. **K-fold** ignores it, diagnostic only. Detectors are deterministic; yfinance and Alpha Vantage match.

Classification, profitability, and the gap.

Three models, two objectives, 100 configurations. The headline finding: F1 and cumulative return disagree.

0.569
BEST F1 · CV

+18.7%
CUM. RET · BAGGING

0.79
WIN RATE · 15 / 19

0.39
SHARPE

MODEL COMPARISON · VALIDATION & TEST

MODEL	VAL $F1_M$	TEST $F1_M$
Random Forest	.393	.162
Bagging	**.425**	**.275**
Baseline · stratified	.361	.447
SPY buy & hold (ref.)	–	+62 %†

Fig 10.1. Bagging wins F1 on validation (.425). On the 28-event test fold the stratified baseline edges ahead on both accuracy and F1, small-sample variance.

Sample-size honesty. 28 test events, 19 trades. Wilson 95 % CI ≈ [0.57, 0.92]. CV distributions are the more reliable signal. † SPY ≈ +62 % buy-and-hold over the same window.

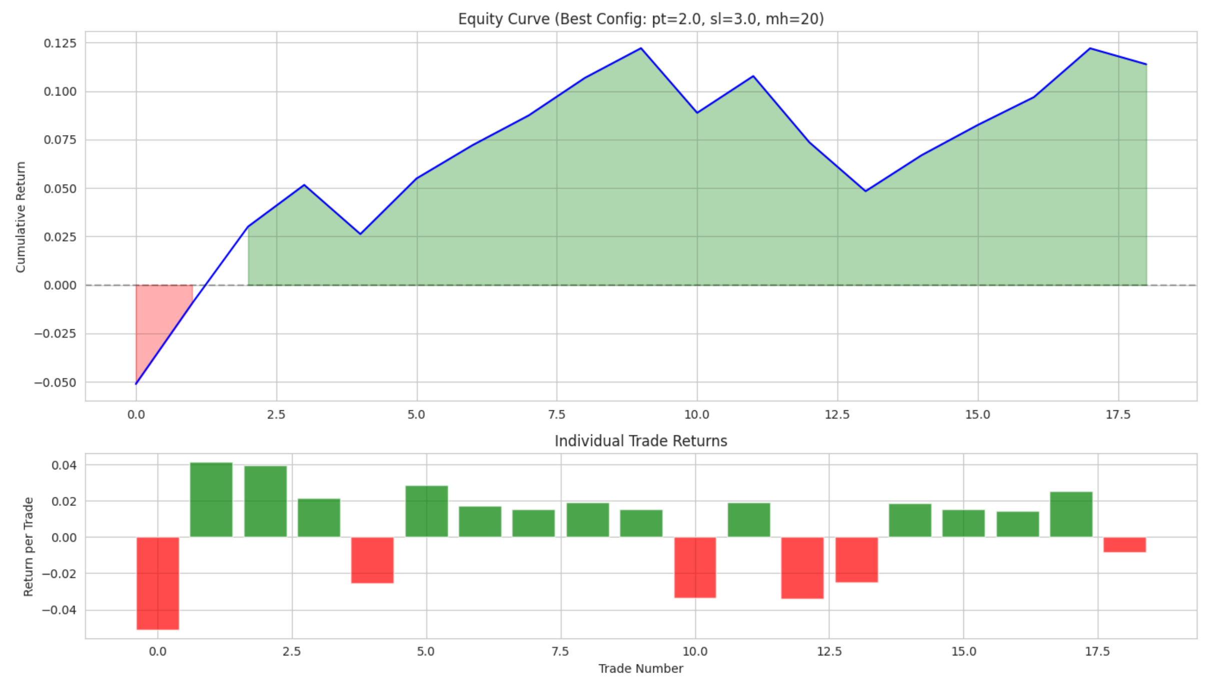

Fig 10.2. Equity curve · profit-optimal config (pt = 2.0, sl = 3.0, mh = 20). **Bagging realises +18.7 %**, Sharpe 0.39, PF 2.30. Grid-search ceiling +28.7 %, the gap is the cost of imperfect classification.
ing +25.85 %, the gap is the cost of imperfect classification.

When F1 and return disagree.

Maximising macro-F1 picks moderate barriers that classify well on small moves. Maximising return picks `pt 2.0 / sl 3.0 / mh 20`, fewer correct labels, each one moves further. **The two optima do not coincide.**

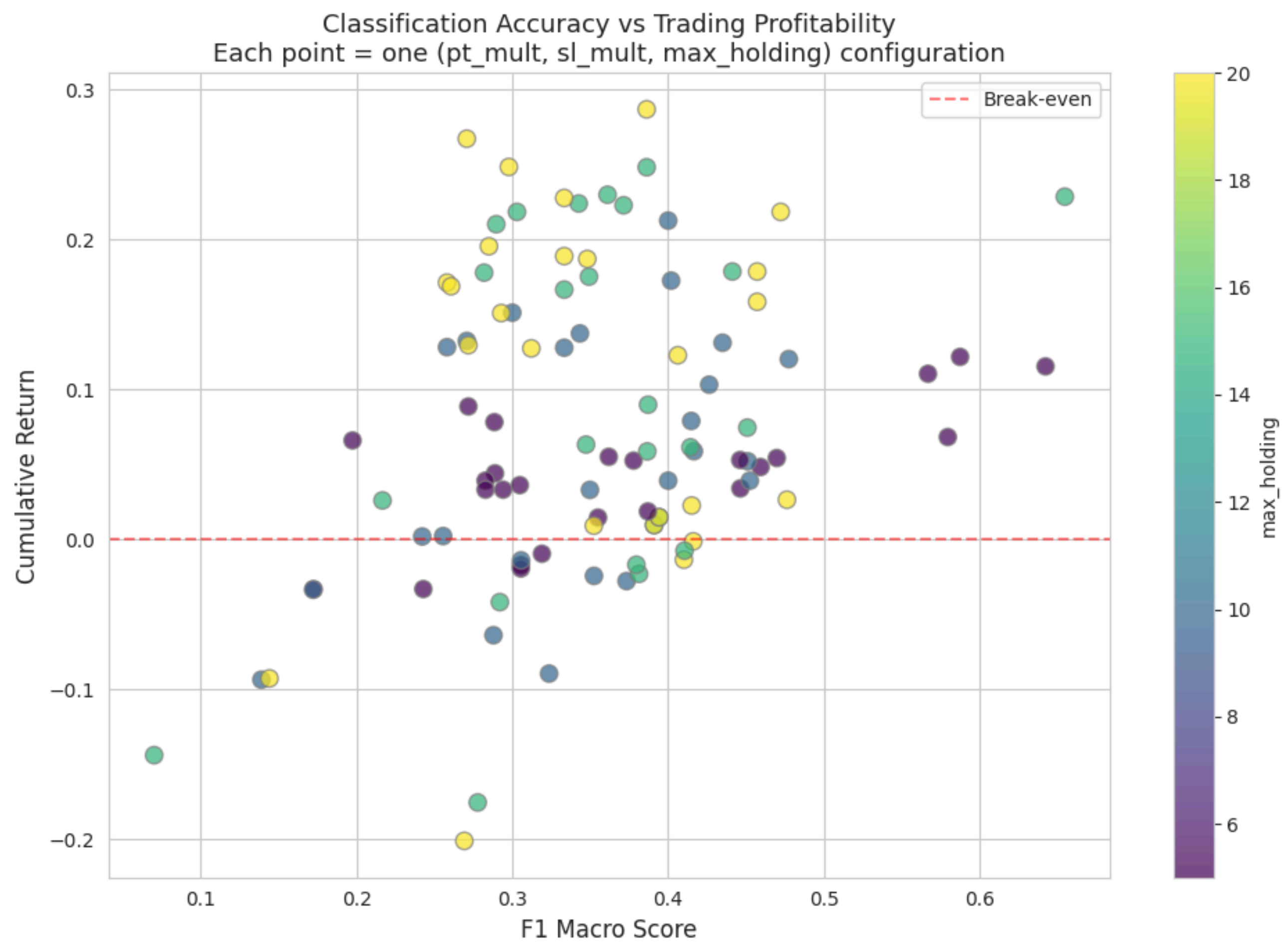

Fig 11.1. Each dot is one of the 100 (pt, sl, mh) configurations. The Pareto frontier lies along the upper-right edge, no single point maximises both objectives.

F-β as a single dial. β < 1 weights precision (don't trade unless sure); β > 1 weights recall (don't miss the move). The thesis exposes the dial; the cost-aware sweep is on the roadmap.

KEY TAKEAWAYS

- ✓ Event-based learning is viable.
- ✓ Volume statistics dominate.
- ✓ F1 ↔ profit gap demands two optima.
- ✓ Bagging > RF on validation.

What's solved, what's next.

Event-based ML, labelled by a triple barrier, is the right framing for this problem. The open question is no longer *does it work?* but *how cheaply does it scale?*

CONCLUSIONS

01 Event-based labelling works.
Geometric completion + triple-barrier outcomes give a clean supervised problem.

02 Bagging is the right ensemble here.
Beats RF on validation F1 (.425 vs .393) and on cumulative return.

03 F1 and return are different objectives.
Reporting both is necessary; reporting only one is misleading.

04 Sample size dominates.
94 events too few for stable test-set superiority; CV optima are the stronger signal.

WHAT WORKED VS WHAT WAS WEAK

WORKED

- Event-only training set
- Triple-barrier labelling
- Bagging best validation F1
- Walk-forward CV

WEAK

- Test-fold variance > signal
- F1 ↔ profit disagreement
- volume_std overweighted

LIMITATIONS

- **Single asset.** SPY only, generalisation unverified.
- **Small sample.** 94 labelled events; 28-event test has wide CIs.
- **No transaction costs modelled.** Slippage, commissions and borrow not yet implemented.
- **Touch-event noise.** Weaker structures lift recall, depress precision.
- **No regime gate.** HMM / vol-state overlay deferred.

ROADMAP

1. Multi-asset extension · FX, futures, crypto
2. HMM regime layer · vol/trend state conditioning
3. Cost-aware F-β tuning
4. Detector probabilities · replace 0/1 with calibrated scores
5. Online retraining · rolling refit at every event close

Notations.

Abbreviations and units used across the deck. For Q&A and offline reading.

MARKET & INDICATORS

SPY	SPDR S&P 500 ETF, U.S. large-cap stocks.
ATR	Average True Range, volatility unit, 14-bar window.
σ_{ATR}	Barrier width unit. pt · ATR = TP distance.
RSI-14	Relative Strength Index, 14-bar momentum oscillator.
MACD	Moving Average Convergence / Divergence (EMA difference).
BB width / %B	Bollinger Band channel width and close position.
OBV	On-Balance Volume, running signed-volume sum.
SMA	Simple Moving Average. Used in sma_N_dist.
OHLCV	Open, High, Low, Close, Volume.

ML, STATISTICS & PERFORMANCE

$F1_m$	Macro F1, harmonic mean of precision and recall.
F-β	F1 with adjustable precision / recall weight.
Gini	Tree split criterion; Δ Gini = feature importance.
ANOVA F	Tests whether a feature mean differs across labels.
CV	Cross-validation; event-CV folds over events.
RF / Bagging	Tree ensembles. RF samples features; Bagging rows.
Triple barrier	First of $\{+pt \cdot \sigma, -sl \cdot \sigma, t + mh\}$ hit.
Sharpe	Excess return / volatility (per-trade in our impl.).
Profit factor	Σ winning trades / |Σ losing trades|.
Max DD	Maximum drawdown from running peak.
HMM	Hidden Markov Model, deferred regime layer.